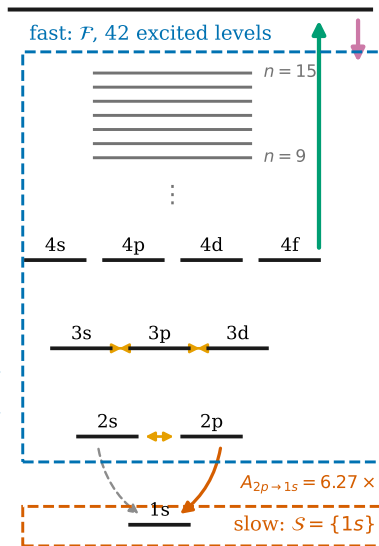


electron impact
excitation and
de-excitation



2s has NO E1 decay to 1s ($A = 0$ exactly). It is still fast:
proton l -mixing to 2p carries 99.97% of its loss rate at the
benchmark, and never less than 99.92% anywhere on the grid. That is
why $S = \{1s\}$ is the only slow state.

schematic: vertical spacing is not the energy scale (see right-hand panel)

continuum ($H^+ + e^-$)

recombination, entering as
the source vector (every state)

$n = 9$ to 15: bundled, 7 states

electron-impact ionisation
(every state)

$n = 5$ to 8: l -resolved, not drawn

$n \leq 8$: l -resolved, 36 states

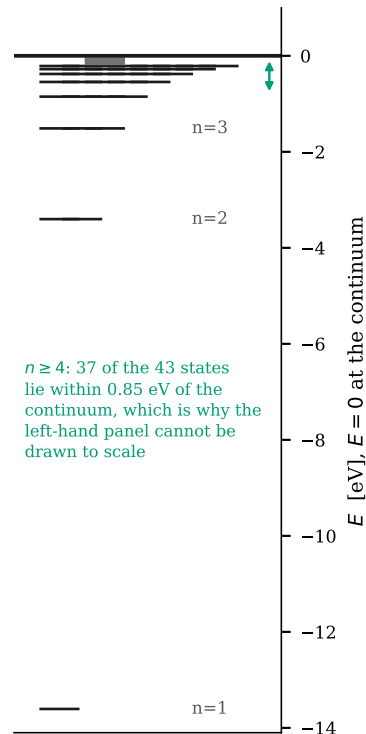
proton l -mixing, $\Delta n = 0$

(35 states: every resolved $n \geq 2$)

$$A_{2p \rightarrow 1s} = 6.27 \times 10^8 \text{ s}^{-1}$$

slow: $S = \{1s\}$

true energy scale



$n \geq 4$: 37 of the 43 states
lie within 0.85 eV of the
continuum, which is why the
left-hand panel cannot be
drawn to scale